

PERSONAL RESEARCH INFRASTRUCTURE

Alloy Lab Database

STAGE 2

The Characterisation Layer

Validated pipelines · Multi-instrument integration · ML-ready schema

PostgreSQL 17

Docker

customtkinter

OpenCV

scipy

numpy

matplotlib

July – September 2026

Experimental data is scattered by default



Instrument files

XRD patterns, VSM loops, SEM images
— on instrument computers, shared
drives, personal laptops.



Spreadsheet analysis

Results manually copied into Origin
projects, Excel files, or text documents
after each measurement.



Implicit connections

"Which XRD pattern belongs to which
sample?" lives in someone's head — or
a sticky note.

"Stage 1 built the data-capture layer. Stage 2 adds the analytical layer — validated characterisation pipelines, stored in a schema designed for machine learning."

STAGE 2 AT A GLANCE

Three instruments. One validated pipeline each.

3

characterisation pipelines

16

database tables · fully populated

170

XRD peaks · fitted & validated

690

SEM images imported

XRD

Peak fitting

NIST LaB₆ validated · lattice param
within 0.0001 Å of certified value

VSM

Multi-segment hysteresis

H_c, M_r, BH_{max}, α/β coefficients, ΔS_m
entropy change · 3 instrument types

SEM

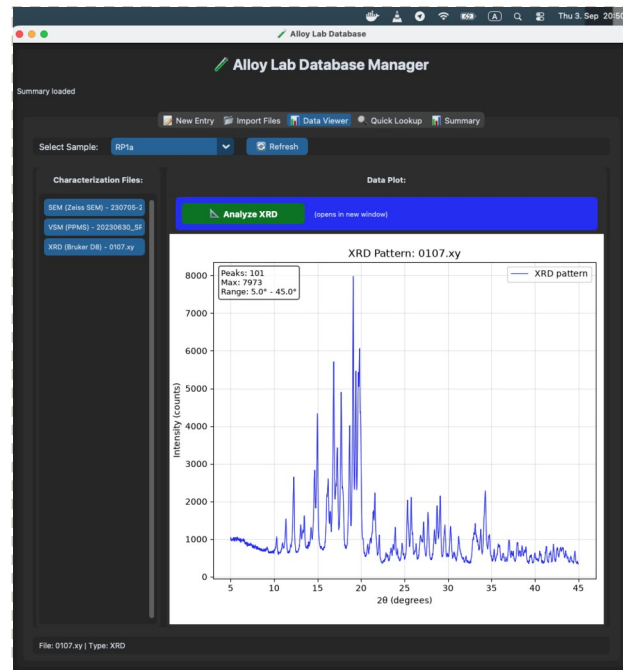
Unified metadata + phase fraction

Zeiss / JEOL / Tescan (16-bit) · operator-
confirmed threshold

X-ray Diffraction — Peak fitting pipeline

✓ Validated against NIST SRM 660c (LaB₆) — lattice parameter within 0.0001 Å of certified value

- Background subtraction and Kα₂ stripping (Cu anode)
- Neighbour-aware peak fitting — each peak uses local context, not isolated windows
- R² quality score per peak — low-quality fits flagged, never silently averaged
- Scherrer crystallite size and d-spacing extracted automatically
- xrd_peaks table: one row per peak (long format)
- xrd_features table: one row per sample, ML-ready wide format



57–58
peaks per sample

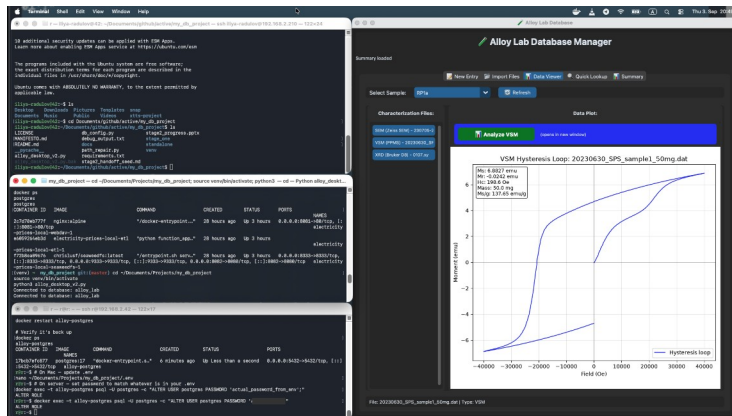
0.938
mean R²

0.0001 Å
vs NIST ref

Vibrating Sample Magnetometry — multi-segment pipeline

Instruments: PPMS VSM · MPMS3 · ACMS | 3 real bugs found and fixed during integration, including a NaN masking issue that silently hid a 60 K temperature jump

- Instrument auto-detection (PPMS/MPMS3/ACMS) — different column conventions per instrument
- Mass extraction from file header (primary) or filename (fallback, with confidence flag)
- Automatic segmentation: MH loops / MT scans / idle / corrupted
- Rolling-window classification — not naive point-to-point derivatives
- Second-quadrant Hc/Mr via crossing-point interpolation on descending branch only
- Temperature coefficients $\alpha(H_c)$ and $\beta(M_r)$ fitted across multiple-temperature series
- Isothermal entropy change ΔS_m — with check for genuine isotherms vs bipolar loops
- Demagnetising correction and BHmax for cuboid samples



Scanning Electron Microscopy — unified metadata + phase fraction

Zeiss

Proprietary compressed binary TIFF tags
(34118/34119)

Pixel size, mag, EHT, detector — parsed from binary string

JEOL

Sidecar .txt file, custom \$KEY value format

Scale bar in pixels + real units → exact pixel calibration. Footer height from \$CM_FULL_SIZE vs actual TIFF height

Tescan

Sidecar .hdr file, standard INI format

PixelSizeX directly in metres. 16-bit image data — handled throughout pipeline

Design decision — why operator judgment is required

Otsu thresholding worked well on one real JEOL image and failed in two different ways on others — texture confusion on a Tescan image (which turned out to be real magnetic domain contrast, confirmed by Kerr microscopy), and noise confusion on Zeiss. A pipeline that pretends to automate a judgment it cannot reliably make is worse than one honest about its limits. The tool suggests Otsu as a starting point; the operator adjusts and saves the confirmed result.

Operator in the loop — Save to DB

Three standalone tools, each launched from the Data Viewer with the file pre-loaded and auto-analysed. Operator reviews, adjusts, and saves confirmed results to the database. No manual transcription.

Data Viewer



Pre-loaded file



Adjust params



 Save to DB

XRD

XRD Analyzer

- Live prominence/distance controls
- Per-peak accept/reject checkboxes
- Mean R^2 recalculated from accepted peaks only
- Save replaces previous analysis for that sample

VSM

VSM MH Analyzer

- Full multi-segment file loaded automatically
- Adjustable branch-detection sensitivity
- Per-segment accept/reject
- Save updates Hc/Mr for confirmed segments

SEM

Phase Fraction

- Otsu threshold as starting suggestion only
- Live slider, optional phase invert
- Calibrated area readout in μm^2
- Save stores operator-confirmed result

16 tables — designed for machine learning from the start

```
alloy_lab (PostgreSQL 17)
├── samples          composition · lineage · VEC/ $\delta$ / $\Delta H$ 
├── compositions     wt% + at% · aimed/measured/ref
├── synthesis         method · temp · atmosphere
├── characterization one row per measurement
├── properties        numeric results per measurement
├── literature_checks MP · OQMD · Alexandria

├── xrd_peaks         one row per fitted peak
├── xrd_features      ML-ready wide format

├── vsm_files         file-level metadata
├── vsm_segments      MH/MT/idle/corrupted
├── vsm_mh_details    Hc · Mr · BHmax · flag
├── vsm_mt_details    + vsm_mt_candidates
├── vsm_temperature_coefficients  $\alpha$  ·  $\beta$ 
```

Feature matrix (X)

- Element fractions wt% + at%
- XRD: peak positions, widths, R^2
- Crystallite size, d-spacing
- Phase fraction (SEM)

Target properties (y)

- Hc, Mr, BHmax
- Temperature coefficients α , β
- Entropy change ΔS_m
- Synthesis route

→ Stage 3 ML

RESULTS

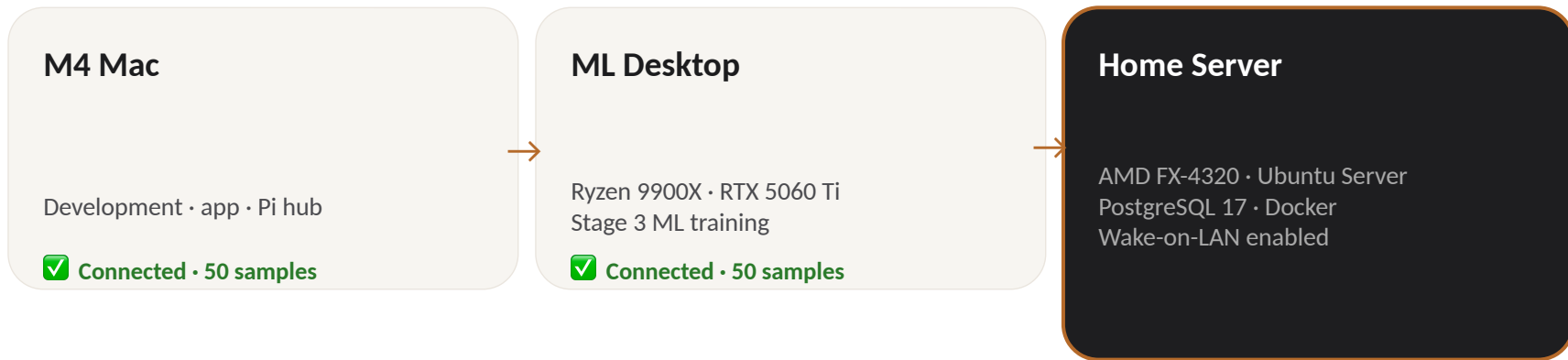
Real data, real validation — anonymised samples

Sample	System	XRD peaks	Mean R ²	VSM Hc	Composition
Sample A	Rare-earth permanent magnet	57	0.938	✓	wt% + at%
Sample B	Rare-earth permanent magnet	58	0.943	—	wt% + at%
Sample C	Rare-earth permanent magnet	55	0.937	—	wt% + at%
Sample D	La-Fe-Si magnetocaloric	—	—	—	wt% + at%
Sample E	Mn-Al-C magnet	—	—	—	wt% + at%
Sample F	Fe-Co-P (synthesis blocked)	—	—	—	wt% + at% + routes

Sample names anonymised pending publication. Alloy systems cover rare-earth permanent magnets, magnetocaloric materials, and novel transition-metal compositions.

Three machines. One canonical database.

On 3 September 2026, the database was migrated from a local Docker container to a dedicated home server. One session, zero data loss.



50
samples

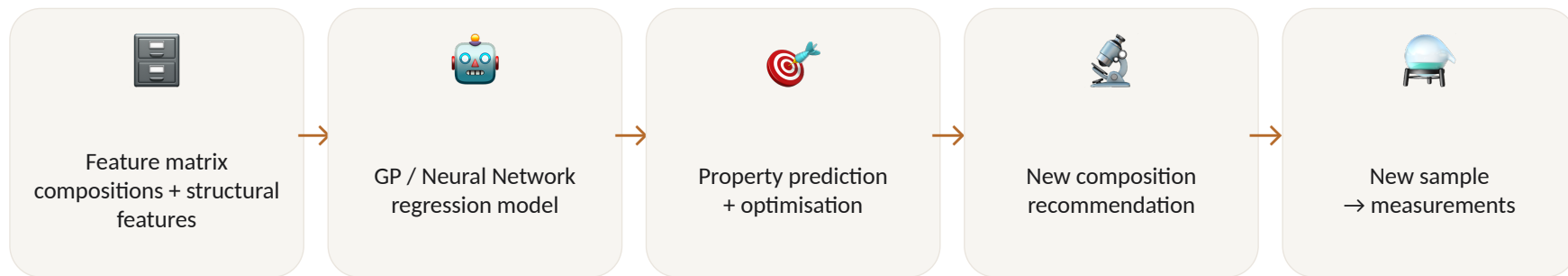
16
tables

170
XRD peaks

92
char. records

The database is ready to train on

Everything built in Stage 2 was designed with this loop in mind:



What is already in the database

compositions

→ Element fractions wt% + at% · aimed / measured / reference

xrd_features

→ n_{peaks} · mean R^2 · crystallite size · d-spacing (ML-ready)

vsm_mh_details

→ H_c · M_r · BH_{max} per segment · quality flags

vsm_temp_coeff

→ $\alpha(H_c)$ · $\beta(M_r)$ — temperature stability coefficients

properties

→ SEM phase fraction · pixel calibration

STAGE 2 • COMPLETE

The hard, instrument-specific work is done.

What remains is science.

✓ XRD pipeline — NIST validated

✓ VSM pipeline — 3 instrument types

✓ SEM pipeline — Zeiss / JEOL / Tescan

✓ Interactive tools with Save to DB

✓ 16-table ML-ready schema

✓ Element-fraction auto-wiring

✓ Synthesis route auto-save

✓ Database on home server

✓ Three machines connected

✓ path_repair.py — clean migration

What started as a way to stop losing track of lab notebooks is turning into a research platform.